

# CLINICAL APPLICATION OF PROTON BEAMS IN THE TREATMENT OF UVEAL MELANOMA: THE FIRST THERAPIES CARRIED OUT IN ITALY AND PRELIMINARY RESULTS (CATANA PROJECT)

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**Background:** The first Italian proton therapy facility was realized in Catania, at the INFN-LNS. With its energy (62 MeV proton beam), it is ideal for the treatment of shallow tumors like those of the ocular region: uveal melanoma, first of all (the most common primary intraocular malignancy of adults) and other less frequent lesions like choroidal hemangioma, conjunctiva melanoma, and eyelid tumors.

**Material and methods:** The first patient was enrolled in February 2002, and to date 30 patients have been treated. All patients had a localized uveal melanoma, with no systemic metastases, and had specific indications for proton beam radiation therapy: lesions between 5-25 mm basal diameter, not exceeding 15 mm thickness, absence of total retinal detachment or glaucoma. According to the tumor dimensions, 2 patients had a small lesion or T1 (6%); 3 had a medium-sized lesion or T2 (10%), 14 had a large lesion or T3 (47%), and 11 had an extra-large lesion or T3 (37%); no patient had extrascleral invasion or T4 of the TNM-AJCC Staging System. In most cases, the tumor infiltrated only the choroid (14 patients, 47%) or the choroid plus the ciliary body (14 patients, 47%). We also treated a primitive iris melanoma, without diffusion to the ciliary body. The target volume was defined as the tumor plus a safety margin of 2.5 mm, laterally and antero-posteriorly; this margin was increased to 3 mm if ciliary body involvement was

present. The treatment was carried out in 4 fractions on 4 consecutive days to a total dose of 54.5 Gy (single fraction 13.6 Gy), which corresponds to 60 CGE (Cobalt Gray Equivalent; single fraction 15 CGE), because the relative biological effectiveness is 1.1.

**Results:** The first follow-up is planned at 6-8 months after the end of the treatment, and our clinical end points are local control (defined as cessation of growth or tumor shrinkage), eye retention, and maintenance of a good visual function. At the time of this writing, we had preliminary results from 13 patients. Nine patients showed tumor shrinkage (69%), 3 a substantially stable dimension (23%), but almost all patients presented an increased ultrasound reflectivity (a surrogate for tumor control).

**Discussion and conclusions:** The literature data show that charged particle therapy has allowed an optimal local control in the treatment of uveal melanomas (about 96% in the different series, superior to that obtained with plaquetherapy [between 83% and 92%]), a metastatic rate slightly better than enucleation reports, and a survival rate of almost 90% at 5 years. Our preliminary results show a tumor response in almost all cases, with no major acute or subacute side effects. We thus plan to continue with our treatment procedures and our dose prescription.

**Key words:** hadron therapy, proton beams, uveal melanoma.

## Introduction

The use of proton beams offers the advantage to improve tumor control, especially for the treatment of small tumors, where it is necessary to obtain a localized dose distribution while sparing the surrounding normal tissues. The hadrons (protons, helium and carbon ions) allow conformation of the dose distribution better than photons or electrons, so the use of these charged particles has developed rapidly in recent years.

There are nearly 20 hadron therapy facilities worldwide, among them about 10 in Europe. In Italy, the first proton therapy facility was realized in Catania, at the Istituto Nazionale di Fisica Nucleare-Laboratori Nazionali del Sud (INFN-LNS). It produces a 62 MeV proton beam, which is ideal for the treatment of shallow tumors like those of the ocular region.

The CATANA project (Centro di Adro Terapia ed Applicazioni Nucleari Avanzate) was developed to treat

ocular pathologies like uveal melanoma, which is the most frequent eye tumor in adults, and in future age-related macular degeneration and other less frequent lesions like retinoblastoma, choroidal hemangioma, conjunctiva melanoma, eyelid tumors, and embryonal sarcoma. Realization of the CATANA project arose from a co-operation between this center and two of the worldwide longest experienced institutions: the Clatterbridge Center for Oncology (CCO-UK) and the Paul Scherrer Institute (PSI, Switzerland)<sup>1-5</sup>.

## Material and methods

### Proton therapy facility

The accelerator in use at the INFN-LNS is a Superconducting Cyclotron that permits a proton beam at an energy between 50 and 70 MeV (average, 62 MeV), with an intensity of 10-20 nA. A dedicated beam line in air has been realized to bring the proton beam to the eye

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of the patient, who is placed in a seated position on the treatment chair (Figure 1). Our treatment chair has 7 independent degrees of movement: 4 movements are related to the chair (forward-backward, right-left, upward-downward, rotation along longitudinal axis), and 3 movements are related to the head fixation system (upward-downward and forward-backward slope, rotation along longitudinal axis). All these movements are under computer control and have a precision in the range of a hundredth of a millimeter. A software case sheet management and automatic setup of the chair has been realized. Moreover, patient fixation is allowed by a thermoplastic mask fixed on a frame and by a bite block in which the patient fits his upper jaw. The final eye position is obtained by the use of two orthogonal X-ray pictures, which allow to verify the tantalum clips position, previously inserted in the eye by the ophthalmologist.

The use of a proton beam in the treatment of ocular pathologies is particularly favorable due to the specific characteristic of the depth dose distribution of the charged particles, called the Bragg-peak phenomenon. It is responsible for an increasing dose deposition of these particles as they travel through the tissue, with a more or less constant low entry dose, a region of high dose (several millimeters wide) at a depth determined by the initial proton energy, and no dose beyond the end of the range (distal falloff). For a 60 MeV proton beam (like the CATANA one), the width of the falloff of the dose from 90% to 10% is about 1 mm at a depth of 3 cm.

A primary Bragg peak like this, with a width of a few millimeters, is not adequate for clinical use because it does not allow complete coverage of the lesion within the treatment isodose. To overcome this obstacle, it is necessary to modulate the Bragg peak itself, and we have done this by the use of two dedicated devices: the modulator and the range shifter, both of them placed inside a specific box along the beam line. The modulator is similar to a propeller with four arms: each modulator has several specific pitches of different thickness, so we

can produce multiple consecutive Bragg peaks at different depths (according to each propeller pitch thickness). The sum of all the resulting peaks produces a modulated Bragg peak or spread-out Bragg peak, with a 100% dose region of a proper width (up to 20-25 mm), allowing the complete coverage of different sized tumors, with the selected margins, within the treatment isodose.

A range shifter is located in the same box where the modulator is placed. It is an essential device that makes it possible to shift the modulated Bragg peak along the antero-posterior axis, according to the necessity to treat anterior or posterior tumors.

### *Clinical applications*

As specified before, the CATANA project has been developed to treat ocular pathologies, especially uveal melanoma in the first phase. Uveal melanoma is the most common primary intraocular malignancy of adults. It arises from melanocytes within the pigmented vascular layer of the eye. The incidence of this pathology is estimated to be between five and seven per million per year, and it is more frequent in fair-skinned people, males and in fifth-sixth decades. Thus, in Italy the incidence is about 300-350 patients per year, but people from other Mediterranean countries could also be treated at the CATANA facility.

Approximately 80% of uveal melanomas involve the choroid and the posterior part of the uvea; less than 10% involve the iris and 10-15% the ciliary body (Figure 2). Depending on their dimensions (largest basal diameter and tumor thickness or height), the AJCC developed a TNM classification for uveal melanomas. An unofficial classification, generally accepted between different researchers, distinguishes 4 groups: small lesion (S, diameter <10 mm and/or thickness  $\leq 3$  mm), medium-sized lesion (M, diameter 10-15 mm and/or thickness 4-5 mm), large lesion (L, diameter 16-20 mm and/or thickness 6-10 mm), extra-large lesion (XL, diameter >20 mm and/or thickness >10 mm).

The prognosis of these lesions depends particularly on: tumor size (the largest tumor dimension has been shown to be the most reliable measure of malignant potential), location of the anterior tumor margin (invasion of the ciliary body correlates with a worse prognosis), and the presence of scleral penetration or optic nerve invasion. The sites of metastatic spread can be the liver (56%), skin (24% as subcutaneous nodules), lung (7%), bone (7%), and CNS (2%)<sup>6</sup>.

The optimal management of uveal melanoma is undefined. Traditionally, it has been treated by removal of the tumor-containing eye, but in the past two decades there has been the development of new approaches that offer the possibility to maintain the eye and in some cases to conserve vision. Most popular by far is radiation therapy, delivered by charged particle beams (proton or heavy ions) or by episcleral radioactive plaques. Selected tumors can be treated by partial eye wall resection or by photocoagulation and laser treatment.

The choice of treatment depends on size and location of

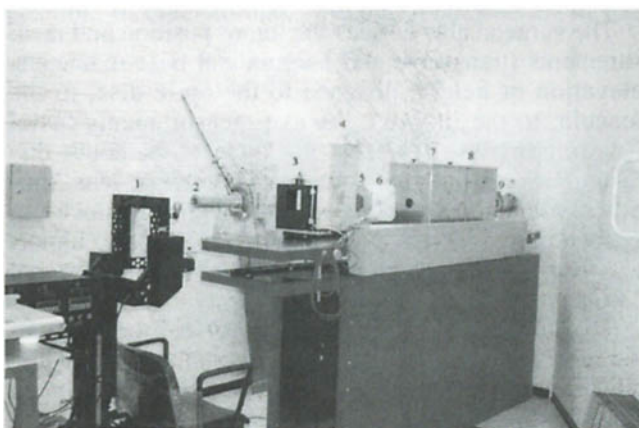


Figure 1 - Beam line and treatment chair (1, chair; 2, final collimator; 3, positioning laser; 4, field light; 5 and 6, monitor chambers; 7 and 8, range shifter and modulator placed in the box; 9, diffusor).

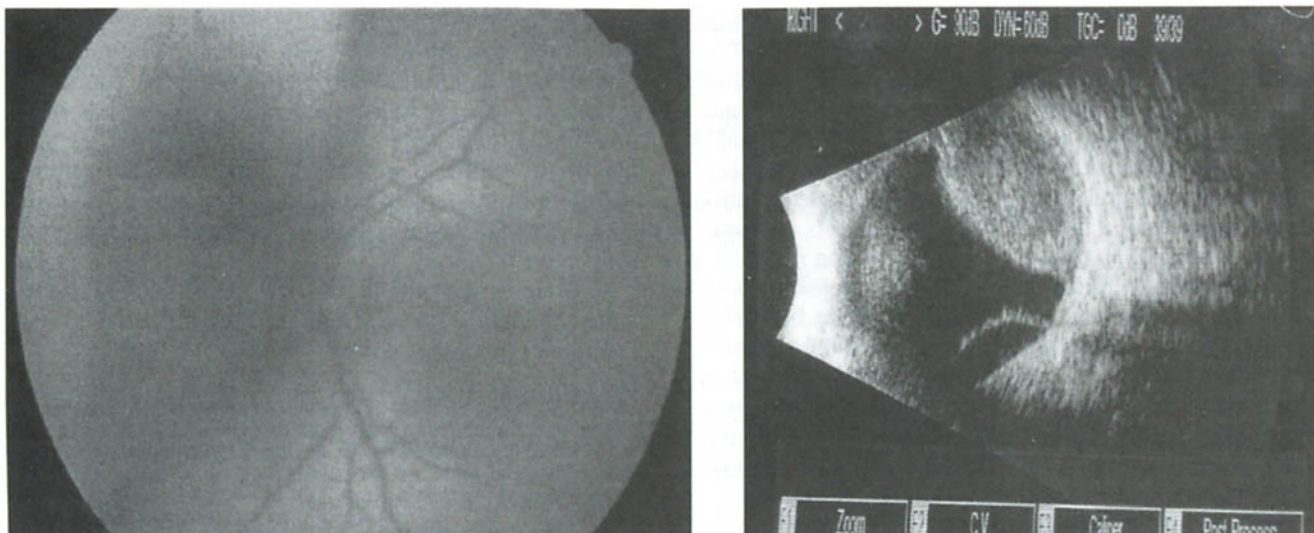


Figure 2 - On the left, a fundus oculi view with a big uveal melanoma localized closed to the optic disk; on the right, the correspondent B-scan ultrasound image shows the elevation of the tumor and the associated retinal detachment.

the tumor. The goal is first to achieve the best possible local control, secondly the preservation of vision, and then the cost effectiveness. Thus, a conservative approach is generally chosen for all except the largest melanomas and eyes with secondary glaucoma, extra-scleral invasion or those which definitely invade the optic nerve. Enucleation may be the best treatment option only in these cases, unless metastatic evaluation reveals spread.

Partial eye-wall resection and photocoagulation are possible only in small-sized tumors, <5 mm thick, whereas plaque therapy can generally be chosen for medium-sized lesions (up to 8-10 mm thick) that are >3 mm distant to the fovea or optic disc. The most commonly used isotopes for plaque therapy are  $^{106}\text{Ru}$  (if thickness is <5 mm) and  $^{125}\text{I}$  (if thickness is between 5-8 mm), whereas  $^{60}\text{Co}$  is now widely abandoned. Iodine plaques have a greater penetration, but they can increase complications and, moreover, are more expensive<sup>7</sup>. The distance from the optic disc is very important due to the technical difficulties to fix the plaque near the optic nerve with the risk of under-dosage. Anterior tumors are generally treated with plaques when possible, because of unacceptable eyelid damage caused by charged particles<sup>8-12</sup>.

Hadrontherapy (proton or helium ions beams) can be used for all tumors, especially those with a basal diameter >5 mm (up to 20-25 mm) or close to the optic disc, in which case it can preserve central vision better than plaque therapy. Larger lesions have an increased risk of recurrence, and the rate of severe side effects of radiation can be very high (large retinal detachment, neovascular glaucoma, radiation retinopathy).

Multitudes of clinical trials have been reported on the use of radiation therapy for the treatment of these tumors. It has prevented melanoma growth or achieved a

tumor shrinkage in over 90% of cases, whereas the rate of metastatic spread is almost the same as with enucleation (or lower, in some cases, due to the possibility of metastatic spread during the surgical manipulation). Seddon *et al.*<sup>13</sup> established that treatment choice has little overall influence on survival in patients with uveal melanomas.

#### Treatment procedures

As specified before, protons allow a high dose deposition in a small deep-seated space, with precise irradiation of the target. The radiotherapist therefore needs to know the exact position of the tumor. To do that, tantalum clips (3-7) are placed around the lesion on the outer sclera by the ophthalmologist, generally very close to the tumor (Figure 3)<sup>15</sup>. The ophthalmological workup is performed at the Ophthalmology Institute of the University of Catania.

The surgeon also defines the tumor position and measurements (transverse and longitudinal base diameters, elevation or height, distance to the optic disc, to the macula, to the limbus), the eye measurements (axial length, transverse diameter, thickness of the coats, distance between anterior cornea and posterior lens, limbus diameter, IPD), clip measurements (distance between clips to limbus, clips to clips and clips to tumor) drawing a precise fundus view model with tumor and clip locations.

The ophthalmologist makes use of A-mode and B-mode ultrasound scan, retinal fluorangiography, wide-angle fundus photographs and, especially, surgical measurements to define all those parameters. He also informs the radiotherapist about the visual acuity of the affected as well as the fellow eye, the presence of exudative retinal detachment, ocular pressure and all other



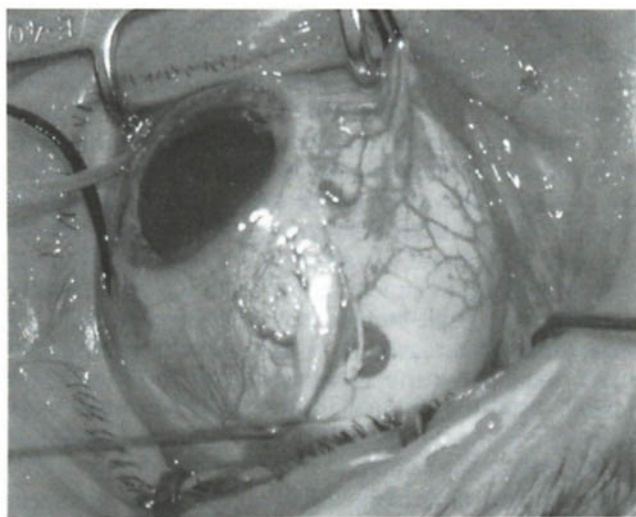


Figure 3 - A tantalum clip sutured around tumor base during the surgical procedures.

information he considers to be significant. A total retinal detachment or the presence of a glaucoma are contraindications to a conservative approach. Together with the diagnosis and the ophthalmological work-up, for all patient a systemic staging is provided through a total body spiral CT scan, to exclude metastatic disease, which is, obviously, another contraindication.

After this first phase the patient arrives at the INFN-LNS, where the workup by the physicist and radiotherapist begins. First of all, a fixation device is made by means of a customized thermoplastic mask and bite block, with the patient in a seated position as specified before. The eyelids are retracted by use of specific retractors or by sticky tape. We prefer the latter because of the psychological impact of the retractors on the patient.

At this point the simulation procedures begin. The patient is invited to gaze at a light point and a pair of orthogonal X-ray pictures (axial and lateral) are taken (Figure 4). This step is repeated 3-6 times with the patient gazing at different points, depending on the tumor position, in order to make the physician sure about the

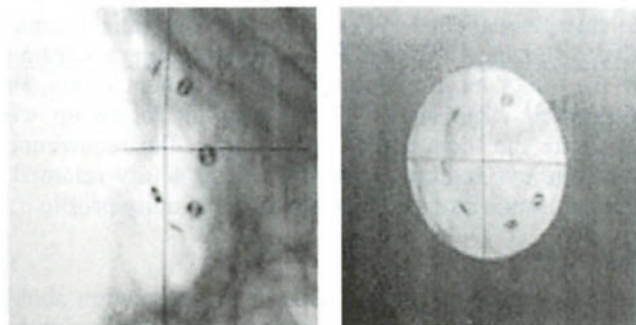


Figure 4 - Axial and lateral X-ray pictures showing the position of the tantalum clips.

position of the clips. A measurement of the eyelid thickness and slope is also needed for the planning procedures.

All the information is then elaborated by means of a dedicated treatment planning system, called EYEPLAN, developed by Goitein and Miller<sup>16</sup> and modified by Perret and Sheen<sup>17</sup>. This software schematically displays a model of the patient's eye, including the lens, optic nerve and fovea, where the tumor is finally drawn by means of the specified measurements and position references (Figure 5).

The target volume is defined as the tumor plus a safety margin of 2.5 mm, laterally and antero-posteriorly, to account for the microscopic peripheral tumor cells, the inaccuracies of the setup and the beam penumbra; this target has to be enclosed within the 90% isodose line. An optimal beam angle and gaze angle are determined to treat the tumor, while sparing as much as possible the optic disc and nerve, macula, ciliary body and lens in descending priority. At the distal and lateral edges of the target, the dose falls from 100% to 0% within 2 mm. A collimator and beam modulator are finally designated the collimator has a thickness of 5 mm and its cross section varies in shape according to that of the tumor to be treated.

A second simulation is performed on the first day of the treatment week, with the eye in the final position to verify and to accept the treatment. If the eyelids cannot be retracted completely outside of the irradiation field, they have to be calculated in the treatment plan.

The treatment is carried out in 4 fractions on 4 consecutive days starting on the day after the second simulation. The prescribed total dose is 54.5 Gy (single frac-

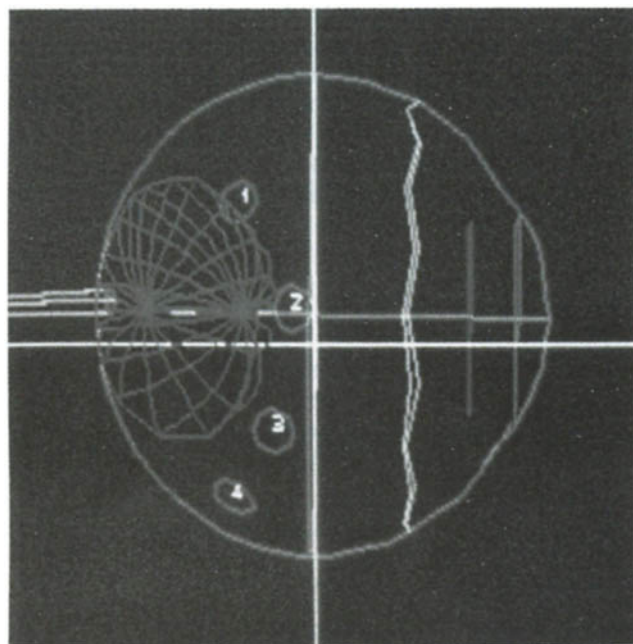


Figure 5 - Lateral view of the eye model created by Eyeplan: the red drawing is the tumor.

tion, 13.6 Gy), which corresponds to 60 CGE (cobalt Gray equivalents; single fraction 15 CGE), because the relative biological effectiveness (RBE) is 1.1. This value of the RBE is currently employed in almost all protontherapy centers worldwide. It is used for all proton energies, for all total doses and dose per fraction, for all tissues, and through all portions of the spread-out Bragg peak.

Before each fraction the eye position is verified and compared to the planned position. A video camera is directed to the eye and allows verification of its position outside the treatment room during the irradiation, which lasts about 30 sec-1 min.

### Patient data

Since February 2002, 30 patients with uveal melanoma, coming from different parts of the country, have been treated at the CATANA facility. All patients had localized disease, with no systemic metastases, and had specific indications for a conservative approach by means of proton beams, depending on their tumor size or location. Among the 30 patients, 19 were women (63%) and 11 men (37%). Patient age ranged from 32-81 years (mean, 54.5). According to the TNM-AJCC staging system, 1 patient was stage I (3%), 4 were stage II (13%), and 25 were stage III (84%). The unofficial classification takes into consideration the same parameters as the TNM, but it differentiates four categories: 2 patients had a small lesion (6%), 3 had a medium-sized lesion (10%), 14 had a large lesion (47%), and 11 had an extra-large lesion (37%).

As regards the localization of the uveal melanoma, 14 tumors invaded the choroid only (47%) and 1 the iris only (3%). Fourteen of the tumors invaded the choroid and the ciliary body (47%). These tumors generally arose from the choroid with a secondary infiltration of the ciliary body, but in 1 patient the tumor was a primitive ciliary body melanoma with infiltration of the choroid periphery. One tumor invaded the choroid, ciliary body and iris as well (3%) (Table 1).

Invasion of the optic disk is the major prognostic factor related to visual outcome. Direct invasion by the tumor

was present in 4 patients (13%), but the number of patients who received a full dose to the optic disk (due to the direct invasion of the disk or to the inclusion of it inside the treatment isodose) was 10 (33%). All these patients will probably become permanently blind. In the same way, patients with lesions with direct invasion of the macula (2 cases) will probably become permanently blind as well, because of the fact that the macula is responsible for the central vision and it is involved in the sharpest vision.

A critical pre-treatment reduction in the visual acuity (less than 3/10, due to the lesion, the secondary retinal detachment, the cataract or to any other pathology which negatively affects visual acuity) was seen in 9 patients (30%). In 3 of them it was so bad that it was necessary to use the healthy eye to fix the red light during the irradiation.

In 17 patients (57%) the lesion caused the development of an exudative retinal detachment, produced by the lesion beneath, and in 3 patient (10%) there was a hemovitreous, treated by means a vitrectomy and silicone oil tamponade. These concomitant ocular pathologies could probably have a major impact on treatment results.

For all patients, the planned tumor volume was defined as the tumor plus a safety margin of 2.5 mm, with the exception of the iris tumor and those tumors arising from the ciliary body. In such cases, a further 0.5-1 mm was added according to the PSI experience<sup>2-5</sup>, which described an increased local control in ciliary body tumors by means of an enlarged planned tumor volume.

All the patients were treated according to the aforementioned procedures, receiving their treatment in 4 fractions of 13.6 Gy (15 CGE, for an RBE of 1.1) to a total dose of 54.5 Gy (60 CGE).

### Results

The most important end point of the conservative approach in the treatment of uveal melanoma with proton beams is local tumor control: it is defined as the cessation of growth or tumor shrinkage. It seems that a persistent tumor mass was not correlated to metastatic potentiality. It takes several months (at least 6-8) to achieve this result, due to the long-term response of melanoma cells (late-responding tissue). It is therefore necessary to wait 6-8 months for a first response from the follow-up. Secondly, the treatment has the objective to save the patient's eye, avoiding the dramatic alternative of enucleation, and thirdly to maintain, according to the tumor location and to the pretreatment vision, an acceptable visual acuity. Thus, during the follow-up, we evaluate the local tumor control rate, the recurrence rate, the eye retention rate, the visual acuity retained, the metastatic rate and, obviously, the toxicity profile.

### Local tumor control

The A-mode ultrasound scan gave information about the ultrasound reflectivity of the tumor, which is dependant on the presence of degenerative phenomena, sclerosis, necrosis and vascular changes<sup>18</sup>. We collected two

**Table 1 - Patient characteristics**

|                                            |                                |                   |
|--------------------------------------------|--------------------------------|-------------------|
| <i>Localization</i>                        |                                |                   |
| Choroid only                               | 14 patients (47%)              |                   |
| Iris only                                  | 1 patient (3%)                 |                   |
| Choroid + ciliary body                     | 14 patients (47%)              |                   |
| Choroid + ciliary body + iris              | 1 patient (3%)                 |                   |
| <i>Tumor dimensions</i>                    |                                |                   |
| Diameter <10 mm and/or thickness ≤3 mm     | Small lesions (S) or T1 (TNM)  | 2 patients (6%)   |
| Diameter 10-15mm and/or thickness 4-5 mm   | Medium-sized lesions (M) or T2 | 3 patients (10%)  |
| Diameter 16-20 mm and/or thickness 6-10 mm | Large lesions (L) or T3        | 14 patients (47%) |
| Diameter >20 mm and/or thickness >10 mm    | Extra large lesions (XL) or T3 | 11 patients (37%) |

values of reflectivity: quantitative and the average, both equally related to the aforementioned phenomena. Starting from the initial values of A-mode reflectivity, an increase in them during the follow-up is suggestive of a radiation-induced degenerative phenomena and, hence, of a tumor response. Moreover, it seems that the amount of the increase is of no prognostic significance. Only an increase in these values, without a dimensional shrinkage of the lesion, is considered evidence of local tumor control.

At the time of this writing, of the 30 patients treated, only the preliminary data at 6 months for the first 13 patients were available. For 11 patients an increase in the A-mode ultrasound reflectivity (quantitative and average values) was detected (85%), and 1 patient showed the same value before treatment and after 6 months (8%). For 1 patient the ultrasound scan could not be used for the presence of silicone oil tamponade after vitrectomy (Table 2).

The dimensional parameters of the tumor are taken by means of ophthalmoscopy and, above all, the B-mode ultrasound scan, especially as regards the elevation (or thickness) of the lesion, whereas the basal tumor diameters are difficult to interpret. Nine tumors showed a reduction in their thickness (69%), 3 tumors were substantially stable in their elevation (23%), and 1 tumor was not assessable (due to the silicone oil) (Table 2). It should be pointed out that the 3 patients who showed a stable elevation of the tumor had an increase in the A-mode reflectivity values, and that the patient unsuitable for ultrasound presented a tumor ophthalmoscopically reduced in its dimensions. Thus, we achieved a local response in almost all cases, but it is too early to say whether it is tumor control or not.

#### *Toxicity and visual outcome*

The use of radiation therapy may cause a variety of acute, subacute and late side effects. In some case, for a neovascular glaucoma or total retinal detachment, enucleation of the eye could be necessary. The evaluation of treatment-related toxicity profiles requires the use of ultrasonography, fluorescein angiography, ophthalmoscopic fundus view and clinical examination.

Of the 13 patients with a preliminary follow-up, 9 presented various degrees of exudative retinal detachment before the treatment session (69%) and 1 had hemovitreous. After 6 months, 6 patients showed a stable retinal detachment (46%), 2 increased (15%), and 1 had a decreased detachment (8%). In 6 (46%) patients a new cataract developed, in 1 (8%) patient of a non-symptomatic neovascular glaucoma, and in 2 (15%) patients of an iridocyclitis. Eyelash loss was detected in 2 patients (15%).

A reduction in visual acuity was seen in 10 patients (77%), but 6 of them (46%) had direct tumor infiltration of the optic disc or macula or those structures were unavoidably included within the treatment isodose. Three patients maintained the same visual acuity as before the treatment (23%) (Table 3).

#### **Discussion**

The use of hadrontherapy is rapidly growing, and there are nearly 20 facilities worldwide that employ proton or heavy-ion beams. The CATANA project is the first in Italy to apply proton beams in clinical oncology, as in the treatment of ocular tumors. Until recently, the approach to uveal melanoma was enucleation, with or without a neoadjuvant conventional radiation therapy. A variety of treatment options have accomplished the need for an eye-preserving therapy: surgical resection, TTT, episcleral plaquetherapy, and stereotactic radiotherapy. However, the most widely used is hadrontherapy, especially by means of proton beams (proton beam radiation therapy).

The application of hadrontherapy in the treatment of uveal melanoma is generally carried out by means of proton beams of 60-70 MeV, an ideal energy to treat tumors to a depth of 3 cm. Several studies have been reported where hadrontherapy has been used to treat uveal melanoma. It is difficult to compare the results of different studies because some series did not randomize patients according to tumor dimensions or type of treatment (whether charged particles or plaque therapy was used).

Clinical studies have reported local control of choroidal melanoma in 80-100% (mean, 95%) of cases,

**Table 2 - Preliminary results**

| Patient no. | A-mode ultrasound reflectivity (quantitative and average values) |                     | Tumor elevation (B-mode ultrasound) |             |
|-------------|------------------------------------------------------------------|---------------------|-------------------------------------|-------------|
|             | Q-A values before PBRT                                           | Q-A values at 6 mos | Before PBRT                         | After 6 mos |
| 1           | 76/48                                                            | 77/69               | 7.3                                 | 7.3         |
| 2           | 93/87                                                            | 93/86               | 2.2                                 | 1.8         |
| 3           | 61/51                                                            | 85/77               | 9.5                                 | 5.5         |
| 4           | 52/41                                                            | 64/66               | 8.5                                 | 6.7         |
| 5           | 46/36                                                            | 73/62               | 5.5                                 | 4.6         |
| 6           | 36/31                                                            | 76/70               | 9.3                                 | 9.2         |
| 7           | 44/35                                                            | 84/75               | 11                                  | 4.5         |
| 8           | 59/63                                                            | 71/61               | 6                                   | 6           |
| 9           | 83/74                                                            | 90/83               | 5.2                                 | 3.7         |
| 10          | 57/48                                                            | 88/84               | 11                                  | 7.5         |
| 11          | 28/20                                                            | 64/45               | 11                                  | 7.5         |
| 12          | 61/51                                                            | 83/77               | 11.8                                | 11.4        |
| 13          | 51/41                                                            | NA                  | 5.8                                 | NA          |

PBRT, proton beam radiation therapy; Q-A, quantitative-average values; NA, not assessable.

**Table 3 - Toxicity and visual outcome**

| Patient no. | Toxicity profile           |                              | Visual acuity |                  |
|-------------|----------------------------|------------------------------|---------------|------------------|
|             | Co-morbidities before PBRT | Side effects at 6 mos        | Before PBRT   | After 6 mos      |
| 1           | Retinal detachment         | Retinal detachment stable    | 10/10         | 6/10             |
| 2           | —                          | —                            | 5/10          | 5/10             |
| 3           | Retinal detachment         | Retinal detachment reduced   | 3/10          | 4/50             |
| 4           | Retinal detachment         | Retinal detachment stable    | 9/10          | 6/10             |
| 5           | Retinal detachment         | Retinal detachment stable    | 2/10          | 2/50             |
| 6           | Cataract                   | Cataract stable              | 3/10          | <i>Motu manu</i> |
|             | Retinal detachment         | Retinal detachment increased |               |                  |
| 7           | Retinal detachment         | Eyelash loss                 | 1/10          | <i>Motu manu</i> |
|             |                            | Conjunctival hypertrophy     |               |                  |
|             |                            | Retinal detachment stable    |               |                  |
| 8           | —                          | Neovascular glaucoma         | 4/50          | <i>Motu manu</i> |
|             |                            | Cataract                     |               |                  |
| 9           | Retinal detachment         | Iridocyclitis                | 7/10          | 7/10             |
|             |                            | Retinal detachment stable    |               |                  |
| 10          | Retinal detachment         | Eyelid hypertrophy           | 6/10          | 3/10             |
|             |                            | Retinal detachment stable    |               |                  |
| 11          | Retinal detachment         | Iridocyclitis                | 6/10          | 1/100            |
|             |                            | Eyelash loss                 |               |                  |
|             |                            | Retinal detachment increased |               |                  |
| 12          | —                          | Cataract                     | 1/50          | (Finger count)   |
|             |                            | Cataract                     |               |                  |
| 13          | Hemovitreal                | —                            | 1/50          | 1/50             |

a recurrence rate of 7.2%, eye retention in almost 85% of patients, and a metastatic rate of 13% (range, 5-20%), favorable compared to those of enucleation series. Almost 50% of patients have maintained a visual acuity equal to or better than 20/200<sup>2,5,8-9,13,19-33</sup>. The results are, obviously, dependent on radiation dose, tumor location and other variables.

Damato *et al.*<sup>34</sup> reported a local control rate of 90-100% and survival rates at 6 years of 100%, 88% and 80% for tumors with the largest tumor dimension of <10 mm, between 10-15 mm and >15 mm, respectively. The ocular conservation rate over 5 years was 90% for small choroidal melanomas and 70% for the largest ones (the reasons for enucleation were: rubeotic glaucoma, retinal detachment and bulky residual tumors). Such results involved more than 2000 patients treated till 1999 with proton beam and Ru plaque therapy (and laser in some cases). The conservation of good visual acuity (6/18 or better) was observed in 50% of small lesions (largest tumor dimension <10 mm) and 23% of large lesions (maculopathy was a common cause of visual loss)<sup>1,34</sup>.

Egger *et al.*<sup>2-5</sup> reported on about 2300 patients treated with proton beam radiation therapy: the 5-year local tumor control probability was improved from 90% (before 1988) to 98.9% (after 1994) after improvements in the technique (increase in the safety margins for tumors located adjacent to the ciliary body, use of the eyelid model developed by Sheen from the CCO in the treatment planning, improvement in the surgical positioning of the tantalum clips)<sup>17</sup>. The 5-year survival rates were 83% and 88%, respectively, for patients treated before and after 1994. Five years after treatment, 89% of patients retained their eye, and at 10 years the rate was 85%.

The PSI seems to report the highest rates of local

control (98.8%), which is the main end point. Gragoudas *et al.*<sup>27-30</sup> obtained a rate of 96.3% using a 160 MeV proton beam and safety margins of 1.5 mm. Char *et al.*<sup>19-22</sup> reached a local control at 5 years of 94.7% using a helium ion beam. Courdi *et al.*<sup>24</sup> reported a local control rate of 89% at 6.5 years (using safety margins of 2.5 mm).

In our series, the short follow-up, about 6-8 months for the first 13 patients treated, has given us only information regarding the tumor response to radiation therapy. Almost all tumors responded to treatment: a shrinkage of the lesion (thickness reduction) was seen in 9 patients (69%) and 3 patients showed a growth cessation with stable disease (23%), whereas an increase in ultrasonographic reflectivity was seen in 11 patients (85%). One patient was evaluated only by means of an ophthalmoscopic fundus view (the silicone oil-filled eye), so the tumor response was detected through clinical criteria. More time is needed to translate these data to an evaluation of local control rate and to report on recurrence rate and survival.

No patient showed major acute or subacute treatment-related side effects. Moreover, the toxicity profile and visual outcome were coherent with literature data.

## Conclusions

Data from the literature show that charged-particle therapy has determined optimal local control in the treatment of uveal melanomas (about 96% in the different series, superior to plaque therapy results, between 83% and 92%), a metastatic rate slightly better than enucleation reports, and a survival rate of almost 90% at 5 years. Different authors agree on the fact that local control and survival rates are clearly related, and that

improvement in patient outcome can be obtained only by improving local control.

Our protocol for dose prescription (54.5 Gy given in 4 fractions of 13.6 Gy on 4 consecutive days) is strongly supported by the literature. In other hadrontherapy centers employing 60-70 MeV proton beams, the total radiation dose used for the treatment of uveal melanoma is almost the same. In addition, our prelimi-

nary results do not show a higher incidence of acute or subacute side effects than other centers. We plan to maintain our treatment procedures and our inclusion criteria: tumors >5 mm but not exceeding 25 mm basal diameter and 15 mm thick, lesions close to the optic disc, absence of total retinal detachment or glaucoma, and need for an eye-preserving therapy (fellow eye blindness).

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